

Question:

John plans to sell all his toys and use the money to buy video games. He has 13 lego sets and he sells them for \$15 each. He ends up buying 8 video games for \$20 each and has \$5 left. How many lego sets does he still have?

Reasoning + Answer:

<tool_call> Reasoning process here :

Step 1 : Calculate the total money John gets from selling the Legos .
He has 13 Lego sets and sells each for \$ 15 .
So , total money from Legos = 13 × 15

Let's compute that :
 $13 \times 15 = (10 \times 15) + (3 \times 15) = 150 + 45 = 195$
So , John gets \$ 195 from selling the Legos .

Step 2 : Calculate the total cost of the video games .
He buys 8 video games at \$ 20 each .
Total cost = $8 \times 20 = 160$

Step 3 : He has \$ 5 left after buying the video games .
So , total money spent = money from Legos - money left
Total spent = $195 - 5 = 190$

But we already know total spent is also 160 (from buying games), which is not matching .

Wait ⚡ contradiction . Let's double-check .

He sells Legos ⚡ gets money ⚡ buys video games ⚡ has \$ 5 left .

So :
Total money from Legos = money spent on games + leftover

Money spent on games = $8 \times 20 = 160$
Leftover = \$ 5
So total money from Legos = $160 + 5 = 165$

But he had 13 Lego sets at \$ 15 each ⚡ $13 \times 15 = 195$

So $195 \neq 165$ ⚡ contradiction .

Therefore , either the problem is inconsistent or we are misinterpreting .

But the problem says : " He has 13 lego sets and he sells them for \$ 15 each " ⚡ so total money is $13 \times 15 = 195$.

Then he buys 8 video games at \$ 20 each ⚡ $8 \times 20 = 160$

He has \$ 5 left ⚡ so total money used = $160 + 5 = 165$

But $195 \neq 165$ ⚡ he should have $195 - 165 = 30$ left , not 5 .

So contradiction .

Therefore , unless he didn't sell all his Legos , the problem is inconsistent .

But the problem says : " John plans to sell all his toys and use the money to buy video games . He has 13 lego sets and he sells them for \$ 15 each ."

So he sells all 13 ⚡ gets $13 \times 15 = 195$.

Buys 8 video games at \$ 20 each ⚡ 160

Leftover = $195 - 160 = 35$ ⚡ but problem says he has \$ 5 left .

So contradiction .

Therefore , either :
- He didn't sell all 13 sets ⚡ but the problem says he sells them ⚡ "sell all his toys" and has 13 sets ⚡ so likely he sells all .

But the problem then asks : " How many lego sets does he still have?"

That implies he may not have sold all .

So perhaps the key is : " he plans to sell all his toys " ⚡ but he may not have sold them all .

But the sentence says : " He has 13 lego sets and he sells them for \$ 15 each " ⚡ that sounds like he sells all .

But then the leftover money doesn't match .

So perhaps the word "them" refers to the sets he sells , and maybe not all?

But it says " he sells them " ⚡ "them" refers to "his toys" , and he has 13 Lego sets .

So likely he sells all .

But the math doesn't work .

So unless the question is a trick ⚡ and the answer is zero ⚡ because he sells all .

But the question is : " How many